Example Solve

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 2xy$$

given that $u = 2$ on $y = x^2$.

- The characteristics of this equation are given by $\frac{dy}{dx} = \frac{B}{A} = \frac{y}{x}$.

We solve this by writing $\frac{dy}{y} = \frac{dx}{x}$, so that $\ln y = \ln x + c$ and $y = xe^c$ or $\frac{y}{x} = p$

where $p = e^c$ is a parameter.

These are straight lines through the origin of arbitrary gradients.

- We also have that

$$\frac{dx}{A} = \frac{dy}{B} = \frac{du}{F}$$

- In this case, $\frac{dx}{x} = \frac{du}{2xy} \Rightarrow dx = \frac{du}{2y}$

- Along the characteristic, $y = px$, and $du = 2y\,dx = 2px\,dx$

- Integrating, we find $u = px^2 + f(p)$, where the "constant" of integration, f, depends on the characteristic we choose and hence on p.
- Eliminating p (using $p = y/x$), we find that the general solution is given by

$$u = xy + f(y/x)$$

- To find f, we use the boundary condition $u = 2$ on $y = x^2$. This implies that

$$2 = x^3 + f(x)$$

$$\text{or} \quad f(x) = 2 - x^3$$

- Then, $f(y/x) = 2 - (y/x)^3$, so that the solution satisfying the boundary conditions is

$$u = xy + 2 - (y/x)^3.$$

Note: the method also works when $F = F(x, y, u)$.

Homogeneous first-order equations

- In a homogeneous equation, $F = 0$.
- Recall that $du = \dfrac{F}{\sqrt{A^2 + B^2}} \, ds$ along the characteristics.

- If $F = 0$, then $du = 0$ also, and u is constant along each characteristic.

<u>Example</u> Solve $2xy \dfrac{\partial u}{\partial x} + (x^2 + y^2) \dfrac{\partial u}{\partial y} = 0$ with the boundary condition $u = e^{x/(x-y)}$ when $x + y = 1$.

- Along the characteristics,

$$\frac{dy}{dx} = \frac{B}{A} \quad \text{so} \quad \frac{dy}{dx} = \frac{(x^2 + y^2)}{2xy}$$

- The RHS contains x and y only in the combination y/x, so we substitute $v = y/x$ (see Mathcentre notes in section 5 on Blackboard).
- Then, $y = vx$, and

$$v + x\frac{dv}{dx} = \frac{x^2 + v^2 x^2}{2vx^2} = \frac{1 + v^2}{2v}$$

so $$x\frac{dv}{dx} = \frac{1 + v^2}{2v} - v = \frac{1 - v^2}{2v}$$

- The equation is then separable, and can be written

$$\frac{2v}{1-v^2}\,dv = \frac{dx}{x} \quad \text{or} \quad \frac{-2v}{1-v^2}\,dv = -\frac{dx}{x}$$

Then,

$$\ln(1-v^2) = -\ln x + c$$

$$1-v^2 = \exp(\ln x^{-1} + c)$$

$$1-v^2 = \frac{e^c}{x}$$

$$= p/x \qquad \text{with } p = e^c$$

so that the characteristics are given by

$$p = x(1-v^2) = x\left(1-\frac{y^2}{x^2}\right) = \frac{x^2-y^2}{x}$$

- For a homogeneous equation, u is constant along each characteristic, so u only depends on p and the general solution of the PDE is

$$u = f(p) = f\left(\frac{x^2-y^2}{x}\right)$$

- To identify the function f, we use the boundary condition. Here,

$$u = e^{x/(x-y)} \quad \text{when } x+y=1 \text{ or } y = 1-x$$

so $$f\left(\frac{x^2-(1-x)^2}{x}\right) = \exp\left(\frac{x}{x-(1-x)}\right)$$

or $$f\left(\frac{2x-1}{x}\right) = \exp\left(\frac{x}{2x-1}\right)$$

and, writing $w = (2x-1)/x$, we see that $f(w) = \exp(1/w)$.

- The solution satisfying the boundary condition is then

$$u = \exp\left(\frac{x}{x^2-y^2}\right)$$

<u>Application</u> : <u>the advection equation</u>

- The <u>advection equation</u>

$$\frac{\partial u}{\partial t} + c\frac{\partial u}{\partial x} = 0$$

is used to model the transfer of one substance (of concentration u) by the flow of another e.g. pollen carried by the wind.
- The characteristics are given by $\frac{dx}{dt} = \frac{B}{A} = c$ so that $x = ct + p$ or $p = x - ct$, where p is a parameter
- Since the PDE is homogeneous, u is constant on each characteristic, so the general solution is

$$u = u(p) = f(x - ct)$$

where f is an arbitrary differentiable function.

- If we have the initial condition $u(x, 0) = \phi(x)$, it follows that $f(x) = \phi(x)$, and the solution becomes $u(x, t) = \phi(x - ct)$
- This is a travelling wave of shape ϕ moving at velocity c.
- To see this, consider $\phi(0)$.
- At $t = 0$, this value occurs at $x = 0$.
- Later, its position is given by $x - ct = 0$ or $x = ct$
- More generally, a given initial value of ϕ moves along a line $x - ct = \text{constant}$.

Characteristic coordinates

- In the previous section, we found characteristic curves $p = p(x, y)$, where p is constant along each curve.
- We solved the differential equation by integrating along the characteristics
- An alternative way of writing this calculation is to transform to a coordinate system based on the characteristics.

Example

- Consider the advection equation

$$\frac{\partial u}{\partial t} + c(x, t)\frac{\partial u}{\partial x} = 0,$$

where the speed now depends on x and t.

- Imagine we have found the characteristic curves $\xi(x, t) = \xi = \text{constant}$ by solving $dx/dt = c(x, t)$.
- We can then write the equation in terms of ξ and $\tau = t$ instead of x and t.

Note: in general, both variables will change, so we use a different letter for the second variable even though $t = \tau$ here.

- These are moving coordinates, and ride with the wave; e.g. in the simple advection equation, we had $\xi = x - ct$.

- The chain rule then gives

$$\begin{aligned} \frac{\partial u}{\partial t} &= \frac{\partial u}{\partial \xi}\frac{\partial \xi}{\partial t} + \frac{\partial u}{\partial \tau}\frac{\partial \tau}{\partial t} \\ &= \frac{\partial u}{\partial \xi}\frac{\partial \xi}{\partial t} + \frac{\partial u}{\partial \tau} \end{aligned}$$

since $\tau = t$, and

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial \xi}\frac{\partial \xi}{\partial x} + \frac{\partial u}{\partial \tau}\frac{\partial \tau}{\partial x}$$
$$= \frac{\partial u}{\partial \xi}\frac{\partial \xi}{\partial x}$$

since τ is independent of x.

- The advection equation then becomes

$$\frac{\partial u}{\partial t} + c(x, t)\frac{\partial u}{\partial x} = \frac{\partial u}{\partial \xi}\frac{\partial \xi}{\partial t} + \frac{\partial u}{\partial \tau} + c(x, t)\frac{\partial u}{\partial \xi}\frac{\partial \xi}{\partial x} = 0$$

$$= \frac{\partial u}{\partial \tau} + \left(\frac{\partial \xi}{\partial t} + \frac{dx}{dt}\frac{\partial \xi}{\partial x}\right)\frac{\partial u}{\partial \xi} = 0$$

- The term in brackets is the total derivative $d\xi / dt$, which is zero since we are differentiating $\xi(x, t) = \xi =$ constant.
- In characteristic coordinates, the advection equation then reduces to

$$\frac{\partial u}{\partial \tau} = 0$$

- This approach can also be applied to second-order equations

Separation of variables

- Applies only to linear PDEs : uses the principle of superposition.
- We suppose that the solution $u(x, y)$ can be written as

$$u(x, y) = X(x)\, Y(y)$$

- Then, the PDE can sometimes be written as two ODEs for X and Y

respectively.

Example Solve $\dfrac{\partial u}{\partial x} - x^2 \dfrac{\partial u}{\partial y} = 2x^2 y u$

given $u(x, 0) = \cosh(x^3)$.

- Writing $u(x, y) = X(x)\, Y(y)$, the PDE becomes

$$X' Y - x^2 X Y' = 2x^2 y X Y$$

- Dividing by $u = XY$,

$$\frac{X'}{X} - x^2 \frac{Y'}{Y} = 2x^2 y \quad \Rightarrow \quad \frac{X'}{X} = 2x^2 y + x^2 \frac{Y'}{Y}$$

or

$$\frac{1}{x^2} \frac{X'}{X} = 2y + \frac{Y'}{Y}$$

where the LHS depends only on x and the RHS depends only on y.

- Since x and y are independent, this means that each side is equal to a <u>separation constant</u>, so that

$$\frac{1}{x^2}\frac{X'}{X} = \alpha \quad \left(\text{or } \frac{1}{x^2}\frac{dX}{dx}\frac{1}{X} = \alpha \right) \quad \text{and} \quad \frac{Y'}{Y} + 2y = \alpha$$

- The equations are separable:

$$\frac{dX}{X} = \alpha x^2 dx \quad \Rightarrow \quad \ln X = \frac{\alpha x^3}{3} + a$$

and $X = \exp(\alpha x^3/3)\, e^a = A \exp(\alpha x^3/3)$

where a and $A = e^a$ are constants.

- Similarly,

$$\frac{dY}{Y} = (\alpha - 2y)\, dy \quad \Rightarrow \quad \ln Y = \alpha y - y^2 + b$$

and $Y = \exp(\alpha y - y^2)\, e^b = B \exp(\alpha y - y^2)$

where b and $B = e^b$ are constants.

- Then, $u(x, y) = X(x)\, Y(y)$

$$= AB\, e^{-y^2} e^{\alpha(x^3/3 + y)} = C e^{-y^2} e^{\alpha(x^3/3 + y)}$$

where $C = AB$.

- By the principle of superposition, the general solution is given by the linear combination

$$u = e^{-y^2} \sum_{\substack{\text{all } C \\ \text{all } \alpha}} C e^{\alpha(x^3/3 + y)}$$

- To fix C and α, we use the boundary conditions.
- Since the sum is a function of $x^3/3 + y$, we can write

$$u = e^{-y^2} F(x^3/3 + y)$$

- We are given that $u(x, 0) = \cosh(x^3)$, so

$$F(x^3/3) = \cosh x^3$$

- Putting $w = x^3/3$,

$$F(w) = \cosh(3w)$$

and $$F(x^3/3 + y) = \cosh(x^3 + 3y)$$

so that the solution satisfying the boundary conditions is

$$u = e^{-y^2} \cosh(x^3 + 3y)$$